

Notes 9

Definition, usage, and examples of the following commands:

grep

- **Definition:**
 - Used to search text in a given file. Grep works in a line by line basis.
- **Usage/Formula:**
 - `grep + option + search criteria + files to display`
- **Example:**
 - Search any line that contains the word dracula.
 - `grep + dracula + ~/Documents/dracula.txt`
 - Search any line that contains the word dracula regardless of the case.
 - `grep + -i + dracula + ~/Documents/dracula.txt`
 - Search for all the lines that do not contain the word 'war'
 - `grep + -v + war + ~/Documents/helloWorld.txt`
 - Search for more than one word per line;
 - `grep + Ewn + '|horror|love|scare' + ~/Documents/dracula.txt`

awk

- **Definition:**
 - Used for processing displaying text. AWk can work with a text file or from standard output. Awk performs operations line by line.
- **Usage/Formula:**
 - `awk + options + {awk command} + file + files to display`
- **Example:**
 - To print the first column of every line of a file:
 - `awk + '{print $1}' + ~/Documents/Csv/cars.csv`
 - To print a file from a given line: (exclude the first 2):
 - `awk + 'NR > 3 + { print }' + /etc/passwd`

sed

- **Definition:**
 - It can search, find and replace, insert and delete. By using SED, you can edit files without opening them.
- **Usage/Formula:**
 - `sed + options + sed script + file`
- **Example:**
 - To print the first column of every line of a file:
 - `awk + '{print $1}' + ~/Documents/Csv/cars.csv`
 - To print a file from a given line: (exclude the first 2):
 - `awk + 'NR > 3 + { print }' + /etc/passwd`

Other examples:

Pipe

- **Definition:** Allows you to redirect the standard output of a command to the standard input of another.
- **Usage:** `command 1 | command 2 | command 3`
- **Example:**
 - To look for a string in a particular man page:
 - `man ls + |grep+ "human-readable|`
 - To display only the 2nd line in a file:
 - `head -2 file.lst + | tail -1`
 - To parse a file with grep and replace a string in the output.
 - `grep -i "honda" cars.csv + |sed 's/honda/tesla/g'`
 - Translate one character to another:
 - `cat file.txt | tr '.' ','`

Standard Output

- **How to save standard output:**
- **Usage:** `command output + > + file`
- **Example:**
 - To save the output of a command to a file:
 - `ls -lA ~ + > + all-files-in-home.txt`
 - To save the error and success to the same file:
 - `ls -lA + downloads/Pictures + > + sucess.txt 2> error.txt`
 - Do not display errors. Send errors to the black hole.
 - `ls -lA + downloads/ 2v> /dev/null`

Appending output to a file

Append means to add more to a file instead of overwriting its content. When we use `>` on a file that already exists and contains data, we overwrite whatever is already inside the file.

- **Example:**
 - In the following command prompt, if `allmyfiles.lst` had any data prior to executing the command, that data will be overwritten by the output of `ls -la`:
 - `ls -la > allmyfiles.lst`
 - If we want to keep data, we must use the following command:
 - `ls -la + >> + allmyfiles.lst`
 - This last command will add the output of `ls -la` to the end of the `allmyfiles.lst` file.